

Analyzing Encounter Duration in the SVH System



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Research Question

How do specific operational variables including **clinical diagnosis, seasonal demand, and department** impact patient **Length of Stay (LOS)**?

Why is reducing LOS important?

Optimize Capacity & Throughput

- Ensure bed availability

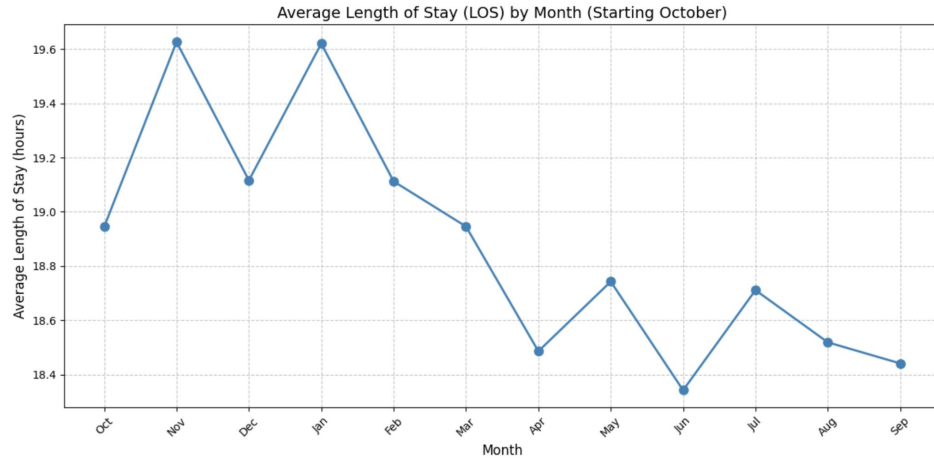
Enhance Patient Safety

- Lower the risk of secondary respiratory infections

Drive Economic Efficiency

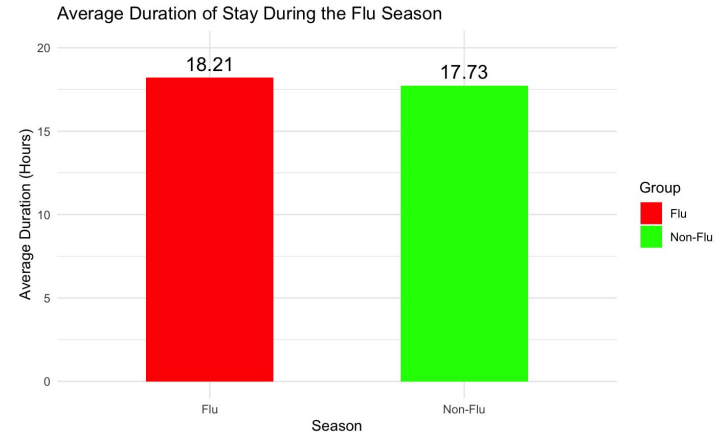
- Reduce the total cost per encounter
- Maximize resource allocation during peak demand

How does Seasonality affect LOS?



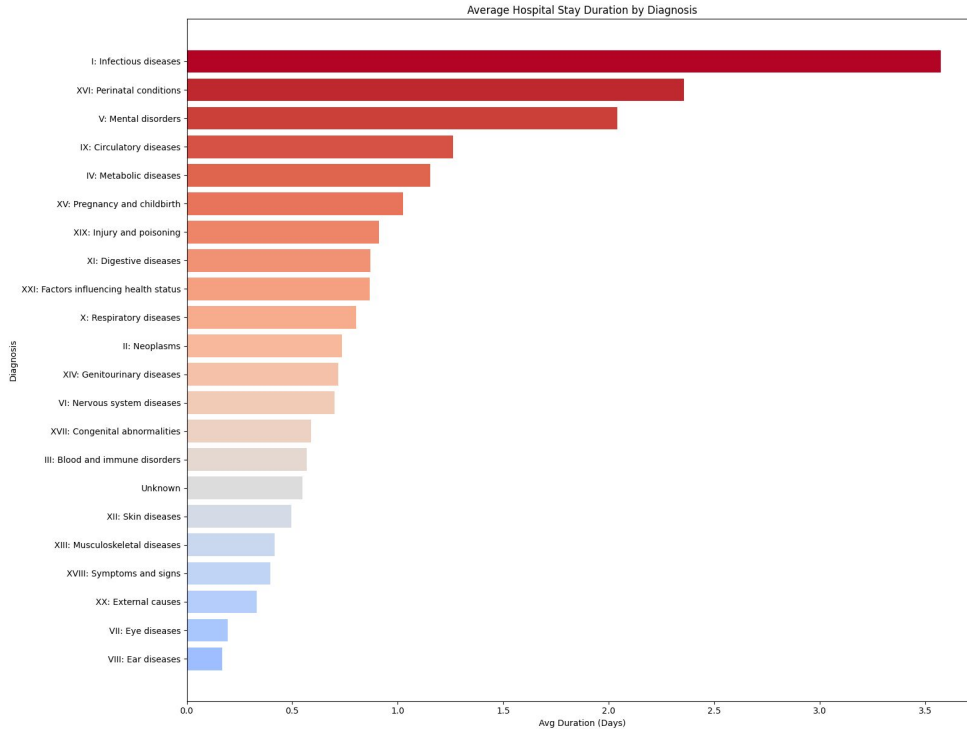
Flu Season

Non-flu Season



Odds ratio: 0.97, $p < 0.001$
Log-link Gamma GLM

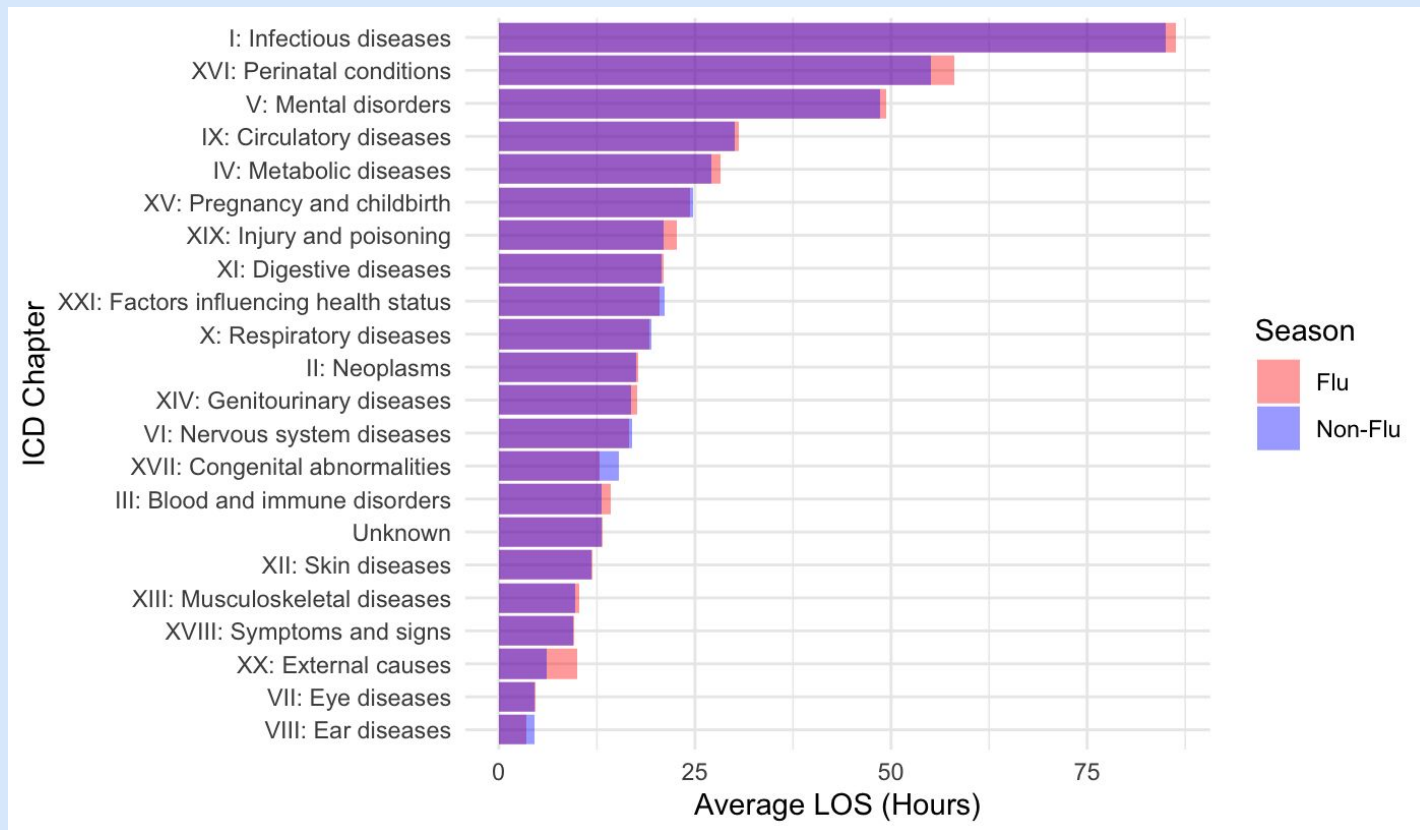
How does Diagnosis affect LOS?



Diagnosis	Avg Length of Stay (Hours)
I: Infectious diseases	85.74
XVI: Perinatal conditions	56.57
V: Mental disorders	49
IX: Circulatory diseases	30.29
IV: Metabolic diseases	27.69
XV: Pregnancy and childbirth	24.59
XIX: Injury and poisoning	21.84
XI: Digestive diseases	20.86
XXI: Factors influencing health status	20.8
X: Respiratory diseases	19.3
II: Neoplasms	17.64
XIV: Genitourinary diseases	17.23
VI: Nervous system diseases	16.84
XVII: Congenital abnormalities	14.15
III: Blood and immune disorders	13.68
Unknown	13.14
XII: Skin diseases	11.89
XIII: Musculoskeletal diseases	10.02
XVIII: Symptoms and signs	9.52
XX: External causes	7.95
VII: Eye diseases	4.64
VIII: Ear diseases	4.03

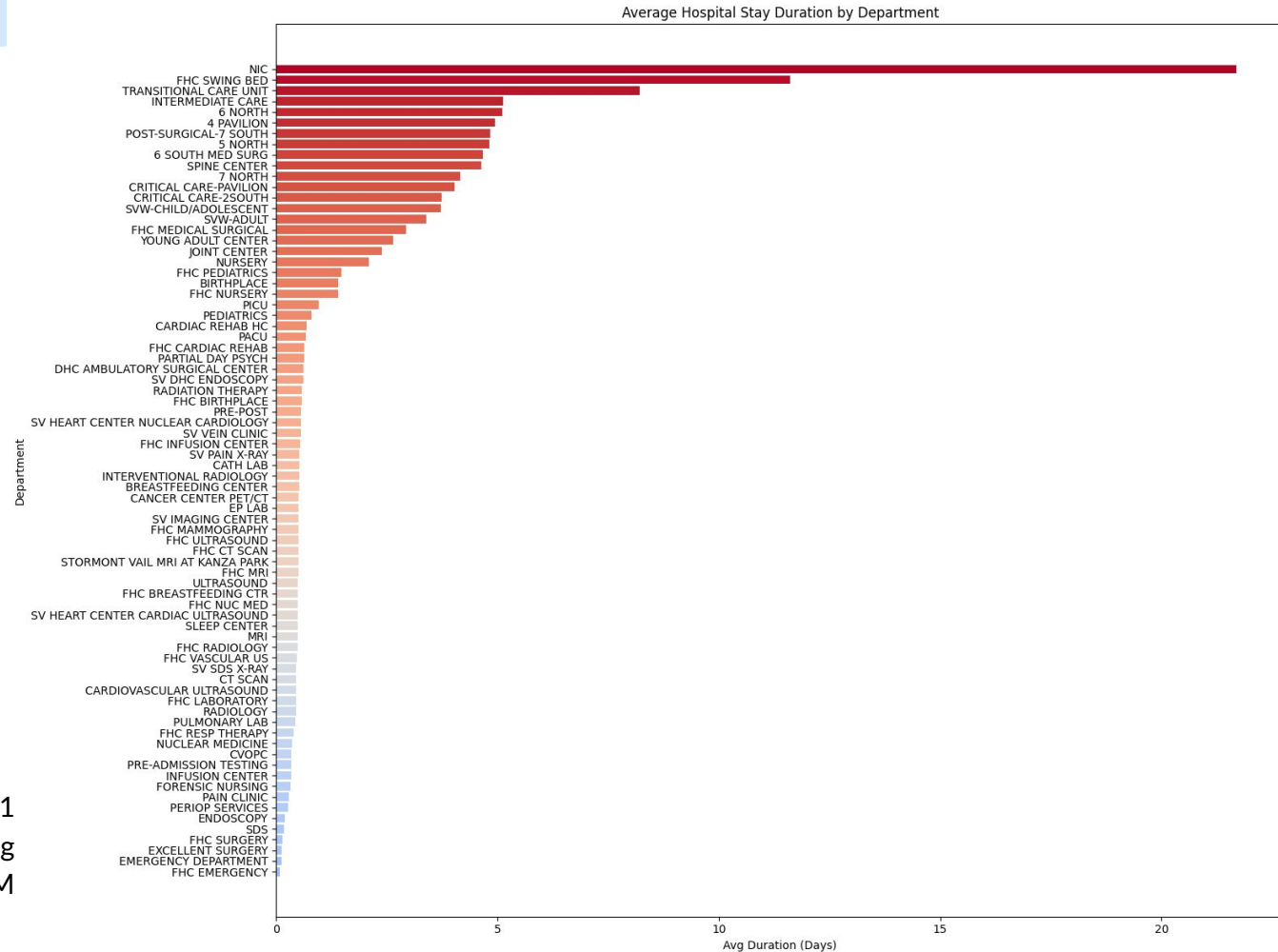
$p < 0.001$
 Analysis of Deviance using
 Log-link Gamma GLM

Diagnosis, Seasonality, and LOS

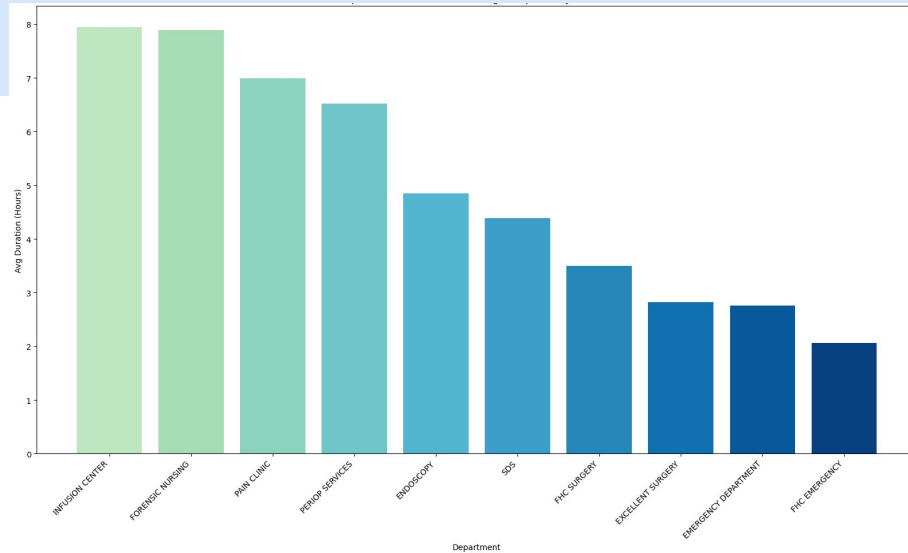
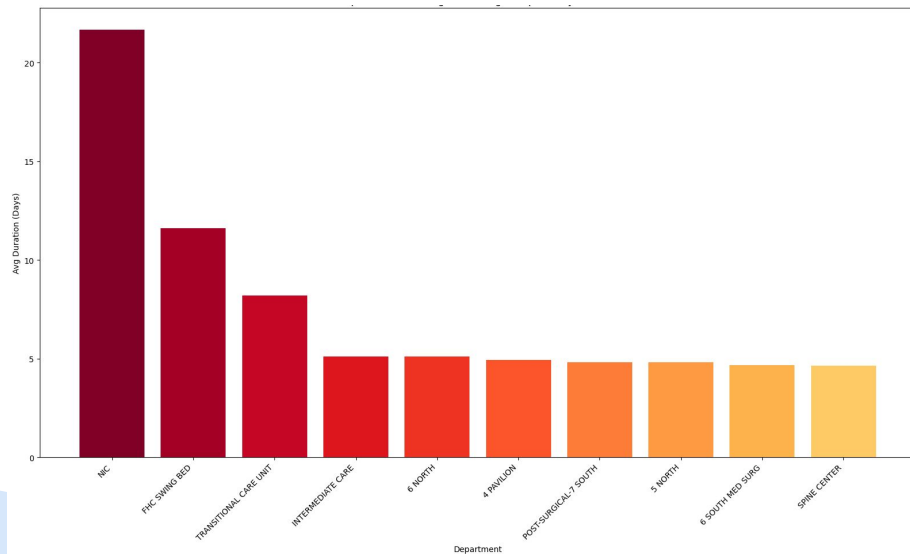


How does department affect LOS?

$p < 0.001$
Analysis of Deviance using
Log-link Gamma GLM



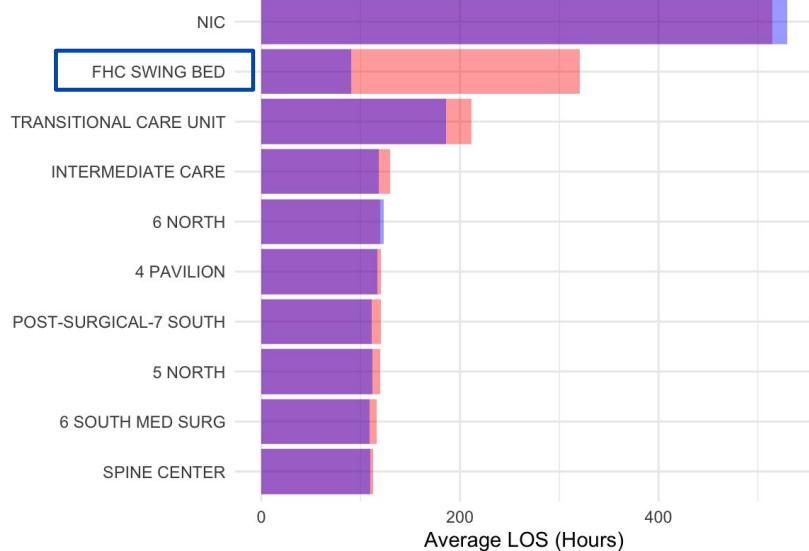
Departments With Highest Average LOS (in days)



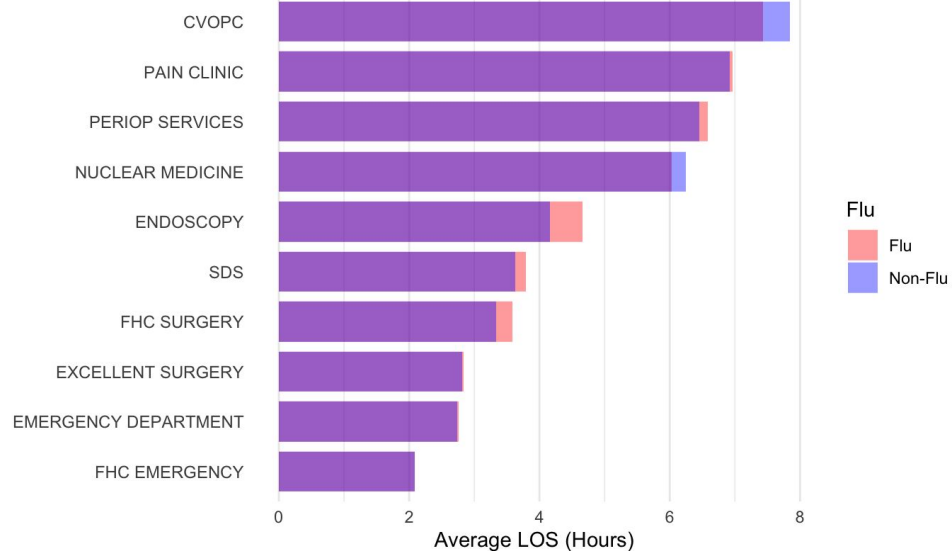
Departments With Lowest Average LOS (in hours)

Department, Seasonality, and LOS

Highest LOS by Department



Lowest LOS by Department



Predictive Models



R ²		Top 5 Features
Model		
Random Forest	0.4172	IsInpatientAdmission, DepartmentKey, AdmitMinute, AdmitDay, AdmitHour
Linear Regression	0.3507	AdmissionSource_Born Outside of this Hospital, AdmissionSource_Federal Hospital, AdmissionSource_TRANSFER FROM ACUTE TO SDU-MEDICARE ONLY, AdmissionSource_Born Outside this Hospital, AdmissionSource_ER VISIT FROM HOME
XGBoost	0.4716	IsInpatientAdmission, AdmissionType_Newborn, DepartmentKey, AdmissionSource_Born Inside this Hospital, IsEdVisit
Single Decision Tree	0.4204	IsInpatientAdmission, DepartmentKey, AdmitHour, IsEdVisit, AdmitMinute

Our Findings!

- **Diagnosis, seasonal demand, and department are all significant predictors of LOS!**
- A patient's average LOS is more likely to increase during flu season
- LOS was elevated during the flu season for 15 out of 22 diagnosis groups
- FHC Swing Bed department has a substantial increase in LOS during flu season